



JEE ADVANCED WEEKEND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: JR.IIT_*CO-SC(MODEL-A)
Time: 3 Hrs

Date: 14-07-24
Max. Marks: 180

14-07-24_JR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_WAT-14_SYLLABUS

- MATHEMATICS:** **PRESENT WEEK (80%): DETERMINANTS:** Introduction of Determinants & its Properties, Properties of Determinants
PREVIOUS WEEK (20%): TRIGONOMETRIC EQUATIONS: General Solution, Solution of Trigonometric equation using graph or otherwise, Equations based on extreme values using graph or otherwise, Trigonometric inequalities using graph or otherwise
- PHYSICS:** **PRESENT WEEK (80%):** Elastic Potential energy of Spring, Spring Block Problems, Stable and unstable equilibrium,
PREVIOUS WEEK (20%): Work, Power, Energy: Work, Kinetic energy, work - energy theorem, conservative and non conservative forces, potential energy, conservation of energy (**EXCLUDE: Problems related to power, Problems related to types of equilibria, vertical circular motion, Spring problems**)
- CHEMISTRY:** **PRESENT WEEK (80%): Atomic Structure:** Dalton's atomic theory, Fundamental Particles - their characteristics; Thomson's model, Rutherford's atomic model, Electromagnetic waves, Plank's Quantum theory & Photo-electric effect, Bohr's atomic model, Applications of Bohr's atomic model - calculation of radius, velocity, frequency, time period and energy of electron in an orbit, Hydrogen spectrum
PREVIOUS WEEK (20%): Liquid Solutions and Colligative Properties: Raoult's law and its application, colligative properties, Abnormal colligative properties & Van't Hoff factor, Abnormal Colligative properties, Henry's law, Different types of distillation, Ideal solution and non ideal solutions, Azeotropic mixtures, DG mixing, Dsmixing

MATHEMATICS :

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – II (Q.N : 9 – 14)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 15 – 18)	Matrix Matching Type	+3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 26)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – II (Q.N : 27 – 32)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 33 – 36)	Matrix Matching Type	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 44)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+3	0	8	24
Sec – II (Q.N : 45 – 50)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 51 – 54)	Matrix Matching Type	+3	-1	4	12
Total				18	60

MATHEMATICS**Max. Marks: 60****SECTION - I****(Maximum Marks : 24)**This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.**Zero Marks:** 0 In all other cases.

1. If $a_1, a_2, a_3, \dots, a_{12}$ are in A.P and $\Delta_1 = \begin{vmatrix} a_1 a_5 & a_1 & a_2 \\ a_2 a_6 & a_2 & a_3 \\ a_3 a_7 & a_3 & a_4 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_2 a_{10} & a_2 & a_3 \\ a_3 a_{11} & a_3 & a_4 \\ a_4 a_{12} & a_4 & a_5 \end{vmatrix}$ then $\Delta_1 : \Delta_2$ is equal to _____

2. If $x > m, y > n, z > r$ such that $(m, n, r > 0)$, $\begin{vmatrix} x & n & r \\ m & y & r \\ m & n & z \end{vmatrix} = 0$, then the greatest value of $\frac{27}{\left(1 - \frac{m}{x}\right)\left(1 - \frac{n}{y}\right)\left(1 - \frac{r}{z}\right)}$ is

3. The remainder when the determinant $\begin{vmatrix} 2017 & -2019 & 2018 \\ -101 & 103 & 117 \\ 1001 & 1003 & -1017 \end{vmatrix}$ is divided by 11 is

4. Let the positive numbers $a_1, a_2, \dots, a_{10}$ be in G.P. & $s, t \in$ set of first 10 prime numbers & $s < t$. Then number of ordered pairs (s, t) such that

$$\begin{vmatrix} \log_7 a_1^s a_2^t & \log_7 a_2^s a_3^t & \log_7 a_3^s a_4^t \\ \log_7 a_4^s a_5^t & \log_7 a_5^s a_6^t & \log_7 a_6^s a_7^t \\ \log_7 a_7^s a_8^t & \log_7 a_8^s a_9^t & \log_7 a_9^s a_{10}^t \end{vmatrix} = 0$$
 is

5. If α, β, γ are different from 1 and the roots of $ax^3 + bx^2 + cx + d = 0$ ($a \neq 0$) and

$$(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) = \frac{15}{2} \text{ and if } \begin{vmatrix} \frac{\alpha}{1-\alpha} & \frac{\beta}{1-\beta} & \frac{\gamma}{1-\gamma} \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} \text{ equals } \frac{pd}{a+b+c+d} \text{ then } [p] \text{ equals } ([.] \text{ is GIF}).$$

6. Let $x, y, z \in \mathbb{R}^+$ and $D = \begin{vmatrix} x & x^3 & x^4 - 1 \\ y & y^3 & y^4 - 1 \\ z & z^3 & z^4 - 1 \end{vmatrix}$. If $x \neq y \neq z$ and x, y, z are in A.P. and $D = 0$ then the value of $2y^2xz + x^2z^2$ is _____

7. Find the number of common solution(s) of the trigonometric equations $\cos 2x + (1 - \sqrt{3}) = (2 - \sqrt{3})\cos x$ and $\sin 3x = 2\sin x$ which satisfy the inequality $\sqrt{3}\tan x - 1 \geq 0$ in $[0, 5\pi]$.

8. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is _____

SECTION – II

(Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

9. Let $a, b > 0$ and $\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$, then which of the following is/are true
- A) $(a+b-x)$ is a factor of Δ
 B) $x^2 + (a+b)x + (a^2 + b^2 + ab)$, is a factor of Δ
 C) $\Delta = 0$ has three real roots (counted as per multiplicity) if $a = b$
 D) $\Delta = 0$ has only one real root if $a = b$
10. The inequality $4\sin 3x + 5 \leq 4\cos 2x + 5\sin x$ is true for Some $x \in$
- A) $\left[-\pi, \frac{3\pi}{2}\right]$ B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ C) $\left[\frac{5\pi}{8}, \frac{7\pi}{6}\right]$ D) $\left[\frac{7\pi}{6}, \frac{13\pi}{6}\right]$
11. The complete set of values of x satisfying $\frac{2\sin 6x}{\sin x - 1} < 0$ and $\sec^2 x - 2\sqrt{2} \tan x < 0$ in $\left(0, \frac{\pi}{2}\right)$ is $(a, b) \cup (c, d)$, then
- A) $a = \frac{\pi}{8}$ B) $b = \frac{\pi}{5}$ C) $c = \frac{\pi}{3}$ D) $d = \frac{5\pi}{12}$
12. Consider the determinant $D = \begin{vmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{vmatrix}$ where x_k are all perfect squares of odd natural numbers. Then which of the following must be TRUE?
- (A) D must be divisible by 16 (B) D must not be divisible by 17
 (C) D may be divisible by 256 (D) D may not be divisible by 64
13. If $\Delta = \begin{vmatrix} x & 2y-z & -z \\ y & 2x-z & -z \\ y & 2y-z & 2x-2y-z \end{vmatrix}$, then
- A) $x-y$ is a factor of Δ B) $(x-y)^2$ is a factor of Δ
 C) $(x-y)^3$ is a factor of Δ D) Δ is independent of z
14. Which of the following is/are true for $\Delta = \begin{vmatrix} a^2 & 1 & a+c \\ 0 & b^2+1 & b+c \\ 0 & b+c & c^2+1 \end{vmatrix}$?
- A) $\Delta \geq 0$ for real values of a, b, c B) $\Delta \leq 0$ for real values of a, b, c
 C) $\Delta = \begin{vmatrix} bc-1 & 0 & 0 \\ 1 & ac & -a \\ -b & -a & ab \end{vmatrix}$ D) $\Delta = 0$ if $bc = 1$, where a, b, c are non-zero.

SECTION - III

(Maximum Marks: 12)

This section contains **FOUR (04)** questions.

Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.

FOUR options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.

For each question, choose the option corresponding to the correct matching.

For each question, choose the option corresponding to the correct matching.

Full Marks : +3 If **ONLY** the option corresponding to the correct matching is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

15. Match the following

(a)	If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$, then $ A+B = ?$	(p)	10
(b)	The maximum value of $\begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$ is	(q)	0
(c)	If a, b, c are in A.P. and $f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}$, then $f'(0)$ is	(r)	-2
(d)	If $\begin{vmatrix} x & 2 & x \\ 1 & x & 6 \\ x & x & x+1 \end{vmatrix} = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$, then a_0 is	(s)	1

The correct option is:-

(A) (I) – R, (II) – Q, (III) – S, (IV) – P

(B) (I) – S, (II) – Q, (III) – P, (IV) – R

(C) (I) – S, (II) – P, (III) – Q, (IV) – S

(D) (I) – P, (II) – P, (III) – S, (IV) – R

16. If α, β, γ are the roots of $x^3 - 3x^2 + 3x - 1 = 0$, then

$$\begin{vmatrix} 1^2 & 2^2 & 3^2 \\ 2^2 & 3^2 & 4^2 \\ 3^2 & 4^2 & 5^2 \end{vmatrix}$$

Then match the list I with list II

List – I		List – II	
(a)	The value of $\alpha\beta\gamma$ is	(p)	0
(b)	The value of $\alpha^2 + \beta^2 + \gamma^2$ is	(q)	3
(c)	The value of $(\alpha-1)(\beta-1)(\gamma-1)$ is	(r)	-8
(d)	The value of $\begin{vmatrix} \alpha-1 & \beta-1 & \gamma-1 \\ \beta-1 & \gamma-1 & \alpha-1 \\ \gamma-1 & \alpha-1 & \beta-1 \end{vmatrix}$ is	(s)	-7

A) (I) – S, (II) – R, (III) – P, (IV) – Q

B) (I) – S, (II) – Q, (III) – R, (IV) – P

C) (I) – R, (II) – P, (III) – Q, (IV) – S

D) (I) – R, (II) – Q, (III) – S, (IV) – P

17. Consider determinant $\Delta = |a_{ij}|$ of order 3. If $\Delta = 2$, then match the following lists:

List – I		List – II	
(I)	The value of determinant $\Delta_1 = (-1)^{i+j}a_{ij} $ is	p)	2
(II)	The value of determinant $\Delta_2 = 2^{i+j}a_{ij} $	q)	2^2
(III)	The value of determinant $\Delta_3 = C_{ij} $, where C_{ij} is the cofactor of a_{ij} is	r)	2^4
(IV)	The value of determinant $\Delta_4 = D_{ij} $, where $D_{ij} = \text{cofactor } C_{ij}$ and C_{ij} is cofactor a_{ij} is	s)	2^{13}

A) I – q, II – p, III – r, IV – s

B) I – p, II – s, III – q, IV – r

C) I – q, II – s, III – p, IV – r

D) I – p, II – s, III – r, IV – r

18. Match the following

List – I		List – II	
(I)	If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$, Then the value of $f(500)$ _____ .	(p)	4
(II)	If $\begin{vmatrix} x & x+y & x+y+z \\ 2x & 3x+2y & 4x+3y+2z \\ 3x & 6x+3y & 10x+6y+3z \end{vmatrix} = 64$, then the real value of x is _____	(q)	0
(III)	Let $D_1 = \begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$ and $D_2 = \begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$ then the value of $\left \frac{D_1}{D_2} \right$, where $b \neq 0$ and $ad \neq bc$, is _____	(r)	8
(IV)	If $x = 31, y = 32$ and $z = 33$, then the value of $\begin{vmatrix} (x^2+1)^2 & (xy+1)^2 & (xz+1)^2 \\ (xy+1)^2 & (y^2+1)^2 & (yz+1)^2 \\ (xz+1)^2 & (yz+1)^2 & (z^2+1)^2 \end{vmatrix}$ is _____	(s)	2

A) I – q, II – r, III – p, IV – s

B) I – r, II – q, III – s, IV – r

C) I – q, II – p, III – r, IV – s

D) I – q, II – p, III – s, IV – r

SECTION - I
(Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

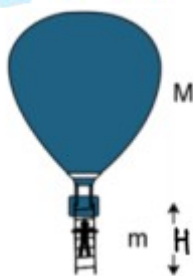
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.

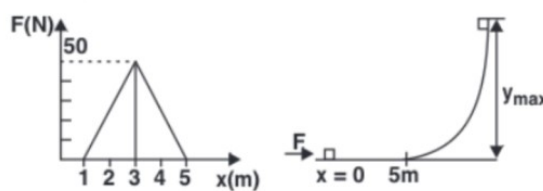
Zero Marks: 0 In all other cases.

19. Mass of balloon + ladder system is M and that of the man is ' m ', Initially, the system is floating in air at rest. 'Man starts climbing up the ladder up to height. "H" assume process to be slow. Find out the work done by pseudo force on man in reference frame of balloon.

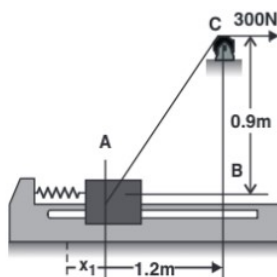


Here "H" is the dist from man to destination from ground. Take $g = 10 \text{ m/s}^2$, $M = 20 \text{ kg}$ & $m = 5 \text{ kg}$.

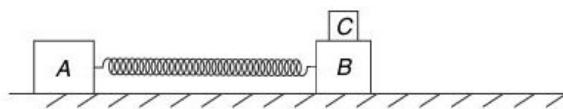
20. A force shown in the $F - x$ graph is applied to a 5 kg cart, which then coast up a ramp as shown. Calculate the maximum height, y_{max} , (in meter) at which cart can reach. ($g = 10 \text{ m/s}^2$).



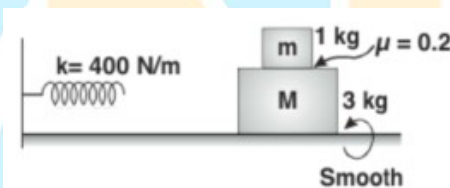
21. The 50 kg block at A is mounted on rollers so that it moves along the fixed horizontal rail with negligible friction under the action of the constant 300 N force in the cable. The block is released from rest at A, with the spring to which it is attached extended an initial amount $x_1 = 0.233 \text{ m}$. The spring has a stiffness $k = 80 \text{ N/m}$. Calculate the velocity (in m/s) of the block as it reaches position B.



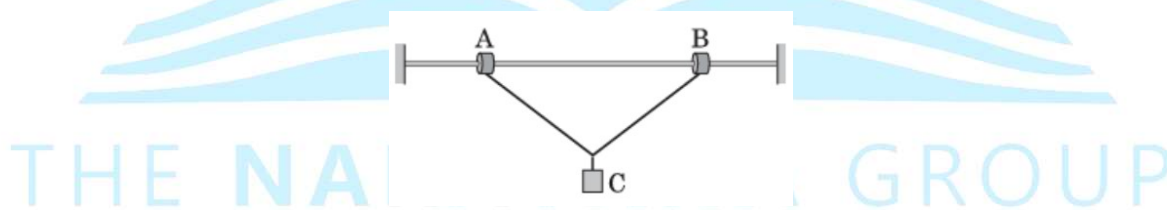
22. Two block A and B are connected to a spring (force constant $k = 480 \text{ N/m}$) and placed on a horizontal surface. Another block C is placed on B. The coefficient of friction between the floor and block A is $\mu_1 = 0.5$, whereas there is no friction between B and the floor. Coefficient of friction between C and B is $\mu_2 = 0.85$. Masses of the blocks are $M_A = 50\text{kg}$; $M_B = 28\text{kg}$ and $M_C = 2\text{kg}$. The system is held at rest with spring compressed by $x_0 = 0.5 \text{ m}$. After the system is released, find the maximum speed of block B during subsequent motion.



23. As shown in the figure, there is no friction between the horizontal surface and the lower block ($M = 3 \text{ kg}$) but friction coefficient between both the blocks is 0.2 . Both the blocks move together with initial speed v towards the spring, compresses it and due to the force exerted by the spring, moves in the reverse direction of the initial motion. What can be the maximum value of v (in cm/s) so that during the motion, there is no slipping between the blocks (use $g = 10 \text{ m/s}^2$).



24. Two identical sleeves A and B each of mass m can slide without friction on a fixed horizontal rod. A load C of mass $2m$ is suspended from the mid – point of a light inextensible cord of length $\ell = 3\sqrt{3} \text{ m}$ connecting the sleeves A and B. Initially the sleeves are at a separation ℓ and the load is held motionless.

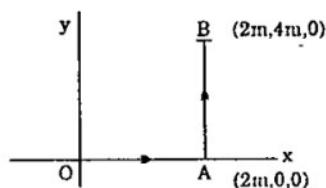


Find out the distance by which load descends when it acquires maximum speed.

25. A stone of mass $m = 5\text{kg}$ is fixed at one end of a uniform flexible chain of linear mass density $\lambda = 0.5\text{kg/m}$. Assembly thus formed is placed on the ground. If the stone is projected vertically upwards with a velocity $v_0 = 20\sqrt{6} \text{ m/s}$, how high will the stone rise. If length of chain is 60 m . Assume air resistance negligible and acceleration due to gravity $g = 10 \text{ m/s}^2$.

[Assume that no mechanical energy is lost as the links of the chain tighten]

26. Find the work done by a force, $\vec{F} = [(y^2 - x^2 + Z^2)\hat{i} + (3xy - 5Z)\hat{j} + (4Z)\hat{k}]$ in taking particle from origin O to the point B (2m, 4m, 0) along the path OAB as shown in the figure.



SECTION – II

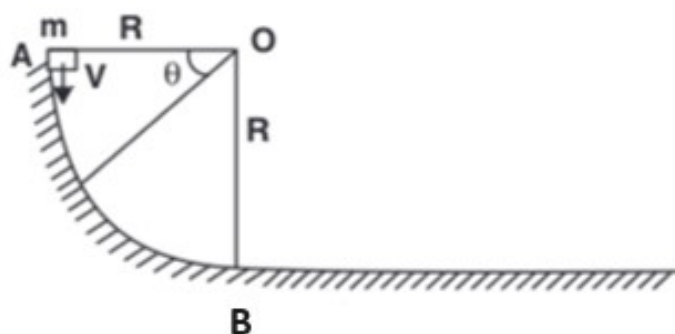
(Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

27. An escalator moves downward at constant speed. You walk up the escalator at this same speed, so that you remain at rest with respect to the ground. Then choose the correct statement.
- A) In the ground frame, work done by man is “Zero”
 - B) In the ground frame, work done by man on Escalator is positive
 - C) The internal energy of man decreases
 - D) In the escalator frame w.d. by man on escalator is positive
28. A particle of mass 1kg moves toward $x = 0$ under the influence of a potential energy function $U(x) = -A|x|^n$, where $A > 0$ and $n > 0$. The particle has barely enough energy to reach $x = 0$. For what values of n will it reach $x = 0$ in a finite time? [Here by barely enough energy we mean that particle reaches origin with zero KE]
- A) 1
 - B) 1.5
 - C) 2
 - D) 2.5
29. A bead, under the influence of gravity, slides along a frictionless wire whose height is given by the function $y(x)$. Assume that at position $(x, y) = (0, 0)$ the wire is horizontal and the bead passes this point with a given speed v_0 to the right. What should the shape of the wire be (that is, what is y as a function of x) so that the horizontal speed remains v_0 at all time? Take $g = 10\text{m/s}^2$
- [y axis is vertically upwards]
- A) $Y = 0$
 - B) $X = 0$
 - C) $Y = -\frac{5x^2}{v_0}$
 - D) $Y = \frac{-5x^4}{3v_0}$
30. There are two massless springs A and B of spring constants k_A and k_B respectively and $k_A > k_B$. If W_A and W_B be denoted as work done on A and work done on B respectively, Then:
- A) If they are compressed to same distance, $W_A > W_B$
 - B) If they are compressed by same force (upto equilibrium state) $W_A < W_B$
 - C) If they are compressed by same force $W_A = W_B$
 - D) If they are compressed by same force (upto equilibrium state) $W_A > W_B$
- [Assume that force is gradually increased until equilibrium is achieved and by “same force” we mean that final value of force is same]

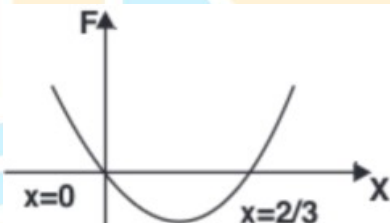
31. Particle 'm' is released with speed $v = \sqrt{gR}$ at point A over a frictional path AB. If speed of particle is constant from A to B then:



- A) μ is variable
 B) $\mu = \frac{\cos \theta}{1 + \sin \theta}$
 C) Work done by friction from A to B is $(-mgR)$
 D) Maximum rate of heat generation due to friction is $mR^{1/2} g^{3/2}$
32. A particle of mass $m = 1$ kg lying on x – axis experiences a force given by law $F = x(3x - 2)$ newton,
 Where x is the x – coordinate of the particle in meters:



- A) The particle is at equilibrium at $x = 0$ and $x = \frac{2}{3}$
 B) The shape of graph of variation of force F (y – axis) with x – coordinate of the particle (x – axis) looks as shown:



- C) The minimum speed to be imparted to the particle placed at $x = 4$ meters such that it reaches the origin is $v = \sqrt{\frac{2600}{27}} \text{ m/s}$.
 D) None of the above

SECTION - III
(Maximum Marks: 12)

This section contains **FOUR (04)** questions.

Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.

FOUR options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.

For each question, choose the option corresponding to the correct matching.

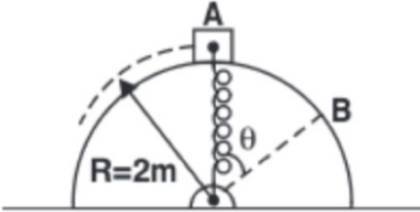
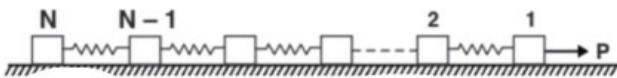
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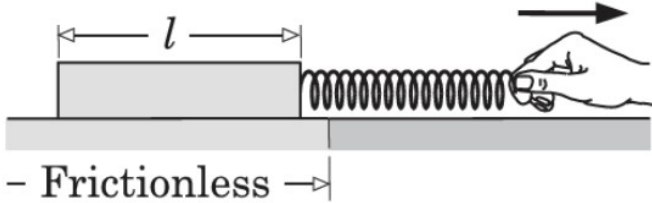
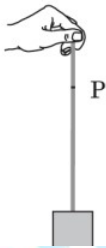
Full Marks : +3 If **ONLY** the option corresponding to the correct matching is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

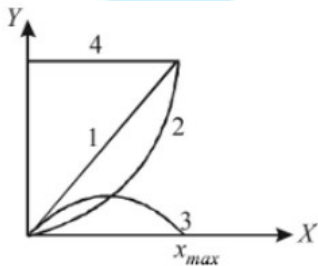
33.

Column – I	Column – II
(i) A 2 kg block is gently pushed from rest at A and it slides down along the fixed smooth circular surface. If the attached spring has a force constant $k = 20\text{N/m}$. What is unstretched length of spring so that it does not allow the block to leave the surface until angle with the vertical is $\theta = 60^\circ$. (radius = 2m)  The diagram shows a block of mass 2 kg at point A on a smooth circular surface of radius $R = 2\text{m}$. A spring is attached to the block and the center of the circle. The block is released from rest at A and slides down to point B, where the angle with the vertical is $\theta = 60^\circ$.	(p) 1.5
(ii) Figure shows N identical blocks connected with identical springs on a smooth horizontal surface. A constant force is pulling the blocks horizontally. During motion all the springs have acquired their stable configuration. If stretch in spring connecting 4th and 5th block is two times that between 8th and 9th block, what is total number of blocks?  The diagram shows N identical blocks connected by identical springs on a smooth horizontal surface. A constant force P is pulling the blocks horizontally. The blocks are numbered 1 to N from right to left. The stretch in the spring connecting the 4th and 5th block is two times that between the 8th and 9th block. If total no. of blocks are N the $\frac{N}{10}$ is.	(q) 1.2
(iii) A portion of the floor in a laboratory is made frictionless. A homogeneous bar of length $l = 1.0\text{m}$ placed on the frictionless portion perpendicular its boundary is being dragged by pulling it with the help of a light spring of force constant $k = 4.0\text{ N/m}$ as shown in the figure. If the pulling agency drags the entire bar out of the frictionless portion by performing minimum amount of work that is $W_{\min} = 17.5\text{J}$ how much energy is lost as heat in this pulling process?	(r) 10

	 - Frictionless - If your ans is N then value of 2N		
(iv)	 One end of an elastic cord is attached to a block of mass m and the other end is held keeping the cord relaxed and vertical. Relaxed length of the cord is l and there is a mark P at a height $0.8l$ from the lower end. If the upper end is slowly raised, the block leaves the ground, when the mark P reaches the point where the upper end of the relaxed cord initially was. Expression for the work done by the force applied by the hand is $\frac{mg\ell}{x}$ and $\frac{x}{5}$ is	(s)	1.6
		(t)	2

- A) i – p, ii – q, iii – r, iv – s
- B) i – q, ii – r, iii – r, iv – s
- C) i – q, ii – r, iii – t, iv – r
- D) i – p, ii – t, iii – s, iv – r

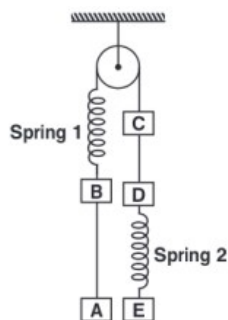
34. A block suspended from a spring is released from rest when spring is unstretched. ‘x’ represents stretch in spring. Select the appropriate graph taking quantities in column I as y – axis.



Column – I		Column – II	
(a)	The KE of block	(p)	1
(b)	The work done on the block by gravity	(q)	2
(c)	The magnitude of work done on the block by spring	(r)	3
(d)	The total mechanical energy of block – earth –s spring system	(s)	4
		(t)	None

- A) a – r, b – p, c – q, d – t
- B) a – r, b – p, c – t, d – s
- C) a – r, b – p, c – q, d – s
- D) a – p, b – r, c – q, d – t

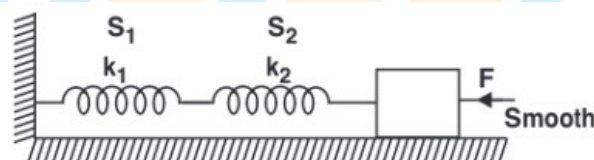
35. The system shown below is in equilibrium. Match the following:



Column – I		Column – II	
(p)	Just after the spring 2 is cut the block D	(i)	Accelerates up
(q)	Just after the spring 2 is cut the block C	(ii)	Accelerates down with $a \neq g$
(r)	Just after the spring 1 is cut the block B	(iii)	Remain at rest
(s)	Just after the string connecting A and B is cut, the block A	(iv)	Moves down with acceleration g .

	P	Q	R	S
A)	1	3	2	4
B)	3	1	1	2
C)	1	3	3	3
D)	1	1	4	4

36. Initially spring are in natural length. An application of external varying force F cause the block to move slowly distance x toward wall on smooth floor.



Column – I		Column – II	
(P)	Work done by S_2 on block	(1)	Zero
(Q)	Work done by S_2 on S_1	(2)	$-\frac{1}{2} \left(\frac{k_1 k_2}{k_1 + k_2} \right) x^2$
(R)	Work done by F on block	(3)	$\frac{1}{2} \frac{k_1 k_2 x^2}{k_1 + k_2}$
(S)	Work done by S_1 on wall	(4)	$\frac{1}{2} \frac{k_1 k_2^2 x^2}{(k_1 + k_2)^2}$

	P	Q	R	S
A)	4	3	1	2
B)	3	2	1	4
C)	1	2	3	4
D)	2	4	3	1

SECTION - I
(Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**

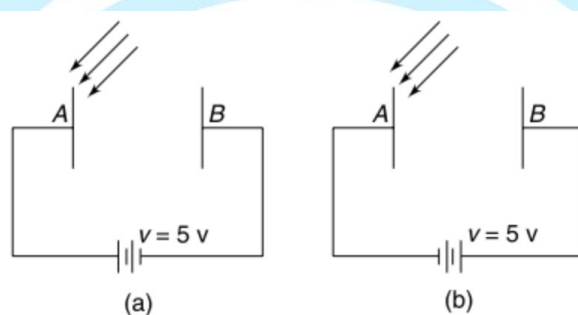
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

37. In a photoelectric experiment, a monochromatic light is incident on the metal plate A. It was observed that with $V=5$ volt, the maximum kinetic energy of photoelectrons striking plate B was 1eV. The polarity of the applied potential difference was as shown in figure (a). With polarity of the applied potential difference reversed (as shown in figure (b)) and frequency of incident light doubled, it was observed that in saturation state, the kinetic energy of electrons striking plate B ranged between 5eV to 20eV. Find the work function of metal (in eV) used in plate A.



38. The difference between the wave number of extreme lines of a spectral series of He^+ ion is $\frac{R_H}{4}$. If the wave number of 2nd line of the series is “p” $\times R_H$, then “p” is
39. A H – like sample has 1st excitation energy of 91.8eV, the ionisation energy of the sample in eV/atom is
40. Pressure over ideal binary liquid mixture containing 10 moles each of liquid A and B is gradually decreased. Given : $P_A^\circ = 100$ mm Hg and $P_B^\circ = 200$ mm Hg, find the pressure in mm Hg which half of the liquid is converted into vapour.
41. A light of wavelength 200nm falls upon a surface and two different wavelength photons $\lambda = 800\text{nm}$ and $\lambda = 400\text{nm}$ are emitted from the surface. 80% of the energy absorbed is re-emitted in the form of photon. Number of photons emitted as $\lambda = 800\text{nm}$ is 3 times that of number of photons emitted as $\lambda = 400\text{nm}$. If the ratio of total absorbed photon to total emitted photon is $x : 1$, then find the numerical value of x.
42. An element with mass number 85 have 4 unit positive charge and 25% more neutrons than electrons in the given ionic state. Its atomic number Z is _____

43. 1.0 mol of H atoms in ground state are excited by irradiating with photons of energy 8.4 eV, 12.09 eV, and 15.0 eV of energy, then the number of spectral lines emitted is equal to
44. The largest wavelength in the ultraviolet region of the hydrogen spectrum is 122 nm. The smallest wavelength in the infrared region of the hydrogen spectrum in nm (round off to the nearest integer) is:

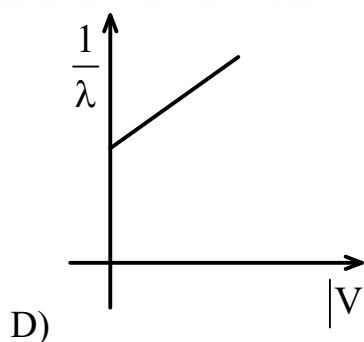
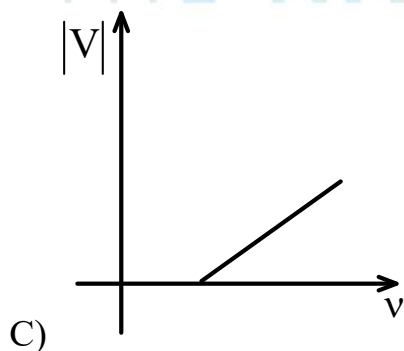
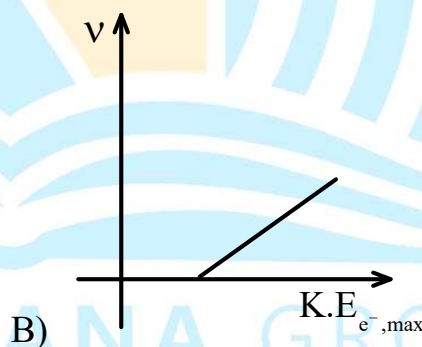
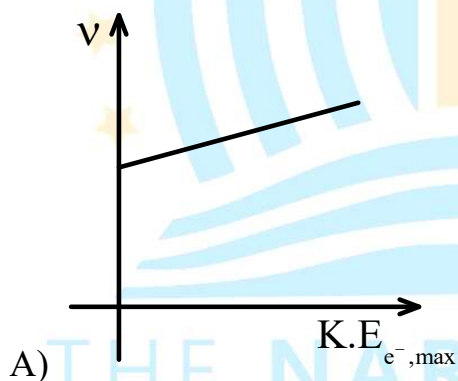
SECTION – II

(Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

45. Photoelectric effect support quantum nature of light because:
- A) There is a minimum frequency of light below which no photoelectrons are emitted
 - B) There is a minimum kinetic energy of photoelectrons depends only on the frequency of light and not on its intensity
 - C) even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately
 - D) electric charge of the photoelectrons is quantized
46. For photo-electric effect experiment, which of the following graphs is correct?
(ν : frequency of incident radiation; λ : wavelength of incident radiation; $K.E._{e^-,max}$: maximum kinetic energy of ejected electron; $|V|$: magnitude of stopping potential)



47. Which is/are correct curve (s):

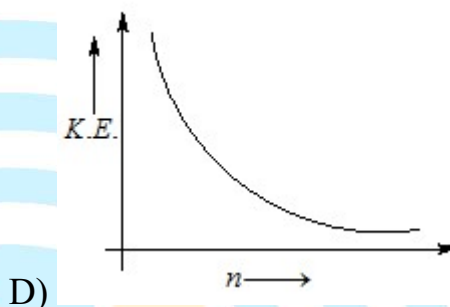
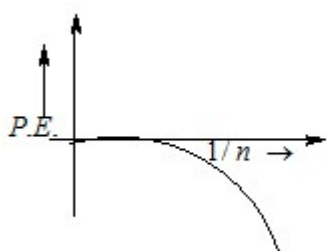
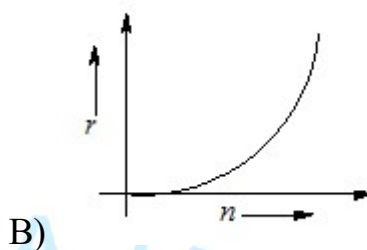
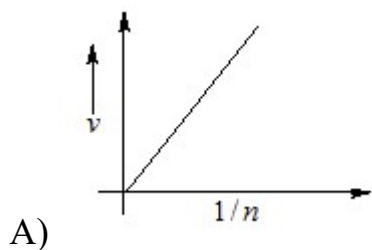
If

v = Velocity of electron in Bohr's orbit

r = Radius of electron in Bohr's orbit

P.E. = Potential energy of electron in Bohr's orbit

K.E. = Kinetic energy of electron in Bohr's orbit



48. In H-atoms sample electrons are de-excited from 4th excited state to ground state. Which is/are correct statements ?

A) No line observed in P-fund series.

B) Total ten lines observed in spectrum.

C) 4 line in UV-region and 3 line in visible region observed.

D) One line observed in Brackett series.

49. Which of the following statements is/are true for anode rays?

A) They are deflected by electric and magnetic fields.

B) Their e/m ratio depends on the gas in the discharge tube used to produce the anode rays.

C) The e/m ratio of anode rays is constant.

D) They are produced by the ionization of the gas in the discharge tube.

50. Two liquids 'A' and 'B' form a solution which shows significant positive deviation from Raoult's law. The normal boiling point of pure liquid 'A' is higher than that of 'B'. Which of the following information(s) related with the solution must be incorrect?

A) Normal boiling point of the solution of any composition will always be less than that of 'A'.

B) Azeotropic mixture of 'A' and 'B' will have normal boiling point lower than that of 'B'.

C) If a small amount of liquid 'B' is added in the liquid solution, the normal boiling point of solution will always decrease.

D) On distillation of the solution at constant pressure of 1 atm, the distillate will have pure liquid 'A' and the residual liquid will be pure liquid 'B'.

SECTION - III

(Maximum Marks: 12)

This section contains **FOUR (04)** questions.

Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.

FOUR options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.

For each question, choose the option corresponding to the correct matching.

For each question, choose the option corresponding to the correct matching.

Full Marks : +3 If **ONLY** the option corresponding to the correct matching is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

51. Match LIST-I with LIST-II, and choose the correct option.

LIST I		LIST II	
I	Positive deviation from Raoult's law	P	Acetone + Chloroform
II	Negative deviation from Raoult's law	Q	Ethanol + Water
III	Negative ΔG_{mix}	R	Carbon di-sulphide + Acetone
IV	Minimum Boiling point Azeotrope	S	Phenol + aniline
		T	water + nitric acid

A) I - R ; II - S ; III - P ; IV - T

B) I - Q ; II - T ; III - R ; IV - Q

C) I - S ; II - P ; III - Q ; IV - R

D) I - P ; II - Q ; III - S ; IV - T

52. According to Bohr's model of H-atom if U_n denotes the potential energy of electron, K_n the kinetic energy, L_n is the orbit angular momentum, v_n is the tangential speed and r_n the radius of nth orbit, then match the following :

List - I		List - II	
I	$\frac{U_n}{K_n} = ?$	p	0
II	$r_n \propto K_n^x, x = ?$	q	-2
III	$\frac{1}{r_n} \propto L_n^y, y = ?$	r	-1
IV	$v_n \propto L_n^z, z = ?$	s	1
		t	2

A) I - q ; II - r ; III - r ; IV - s

B) I - q ; II - s ; III - r ; IV - s

C) I - q ; II - r ; III - q ; IV - r

D) I - q ; II - r ; III - s ; IV - r

53. Match LIST-I with LIST-II, and choose the correct option.

List –I		List – II	
I	0.01M $\text{Al}_2(\text{SO}_4)_3$	P	Solution with highest boiling point
II	0.01 M AlPO_4	Q	Van't Hoff factor is greater than 1
III	0.01 M urea	R	Solution with lowest osmotic pressure
IV	0.01 M MgCl_2	S	Solution with lowest freezing point
		T	Solution with highest vapour pressure

- A) I - P ; II – Q ; III – R ; IV - S B) I – S ; II - R ; III - T ; IV - Q
 C) I -Q ; II – R ; III – S ; IV – P D) I -P ; II - Q; III - T; IV -Q

54. If the shortest wavelength of spectral line of H – atom in Lyman series is x, then match the following for Li^{2+} .

List – I		List – II	
I	Shortest wavelength in Lyman series	P	$\frac{4x}{5}$
II	Longest wavelength in Lyman series	Q	$\frac{4x}{9}$
III	Shortest wavelength in Balmer series	R	$\frac{x}{9}$
IV	Longest wavelength in Balmer series	S	$\frac{4x}{27}$
		T	$\frac{9x}{4}$

- A) I - R ; II – S ; III – T ; IV - P B) I – S ; II - R ; III - T ; IV - Q
 C) I -Q ; II – R ; III – S ; IV – P D) I -R ; II - S; III - Q; IV -P